Deep Learning Project

Topic: Deep Learning-Based Predictive Maintenance Framework for Remaining Useful Life Prediction and Maintenance Decision Support

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Introduction

- Modern industrial systems such as aircraft engines, turbines, and manufacturing equipment rely on continuous monitoring to ensure reliable operation
- Failures in these systems can lead to safety risks, operational downtime, and significant economic losses.
- To address this, industries are increasingly adopting Predictive Maintenance strategies that use sensor data and machine learning to anticipate equipment failures
- A key component of predictive maintenance is Remaining Useful Life (RUL) prediction, which estimates how long a system can operate before failure
- Advances in deep learning models such as CNN and LSTM have significantly improved the ability to analyze complex sensor data and predict equipment degradation patterns

Problem Statement

- Modern industrial systems (e.g., aircraft engines) operate under complex and dynamic conditions, making failures difficult to predict.
- Traditional maintenance strategies (like scheduled or reactive maintenance) are inefficient, costly, unable to prevent unexpected failures.
- Existing data-driven approaches mainly focus on:
  - Predicting Remaining Useful Life (RUL)
  - But do NOT integrate maintenance decision-making

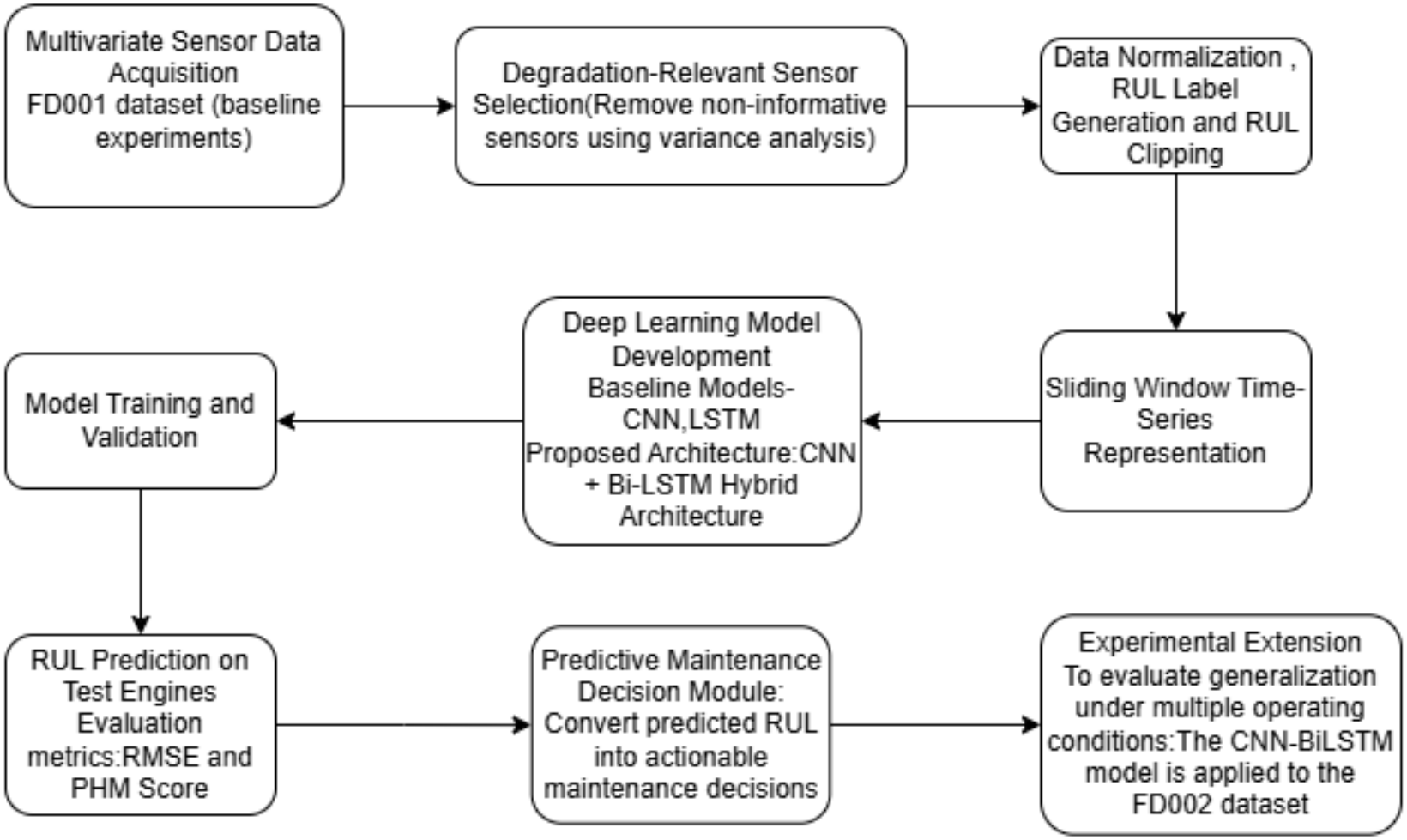
Dataset Description

- The dataset used in this study is the C-MAPSS Turbofan Engine Degradation Simulation Dataset developed by NASA.
- The FD001 dataset consists of three files:
  - train_FD001.txt – Engine run-to-failure data used for model training
  - test_FD001.txt – Engine operational data used for testing
  - RUL_FD001.txt – True Remaining Useful Life values for test engines.
- Each record in the dataset contains:
  - Engine ID
  - Cycle number(time step of operation)
  - 3 operational settings
  - 21 sensor measurements
- Dataset Statistics:
  - 100 turbofan engines in the training set
  - 100 engines in the test set

Project Objective

- To develop a deep learning-based predictive maintenance framework that predicts the Remaining Useful Life (RUL) of a system and supports data-driven maintenance decision-making under real-world operating conditions.
- To evaluate the performance of deep learning models using the NASA C-MAPSS FD001 and FD002 dataset.

Methodology



FD001 Vs FD002

- FD001 has 100 training engines and 100 test engines; FD002 has 260 training engines and 259 test engines.
- FD001 operates under a single operating condition; FD002 operates under 6 distinct flight envelope conditions (altitude, Mach number, throttle resolver angle combinations).
- In FD002 sensor readings show multimodal distributions — 6 operating conditions create 6 distinct clusters per sensor, making degradation signal extraction significantly harder

Experimental Results

- The CNN, LSTM, and CNN-BiLSTM models were implemented for Remaining Useful Life (RUL) prediction using the NASA C-MAPSS FD001 dataset and the evaluation of these models is extended to FD002 dataset.
- Performance was evaluated using:
  - RMSE (Root Mean Square Error)
  - PHM(Prognostics and Health Management) Score metric
- Training and validation curves were analyzed to assess model learning behavior and overfitting.

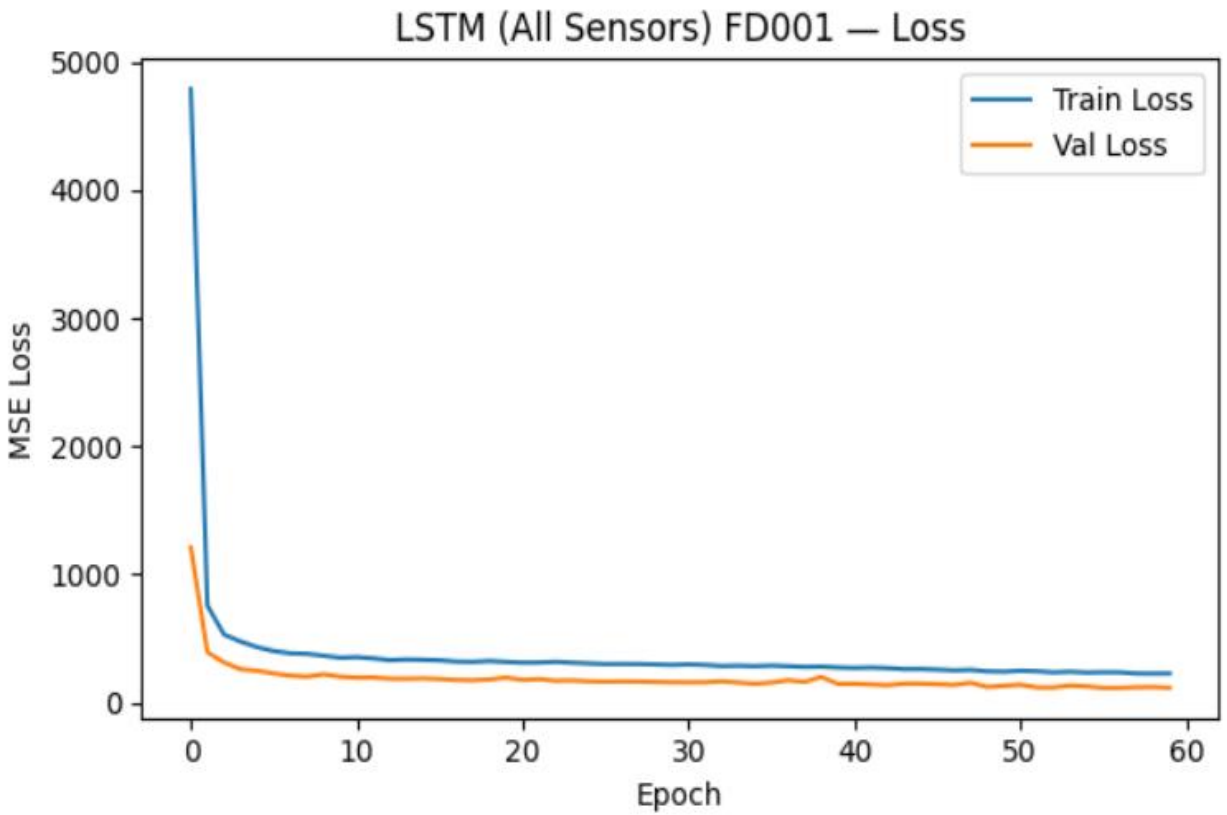
Ablation Study results(Filtered vs non filtered)-FD001

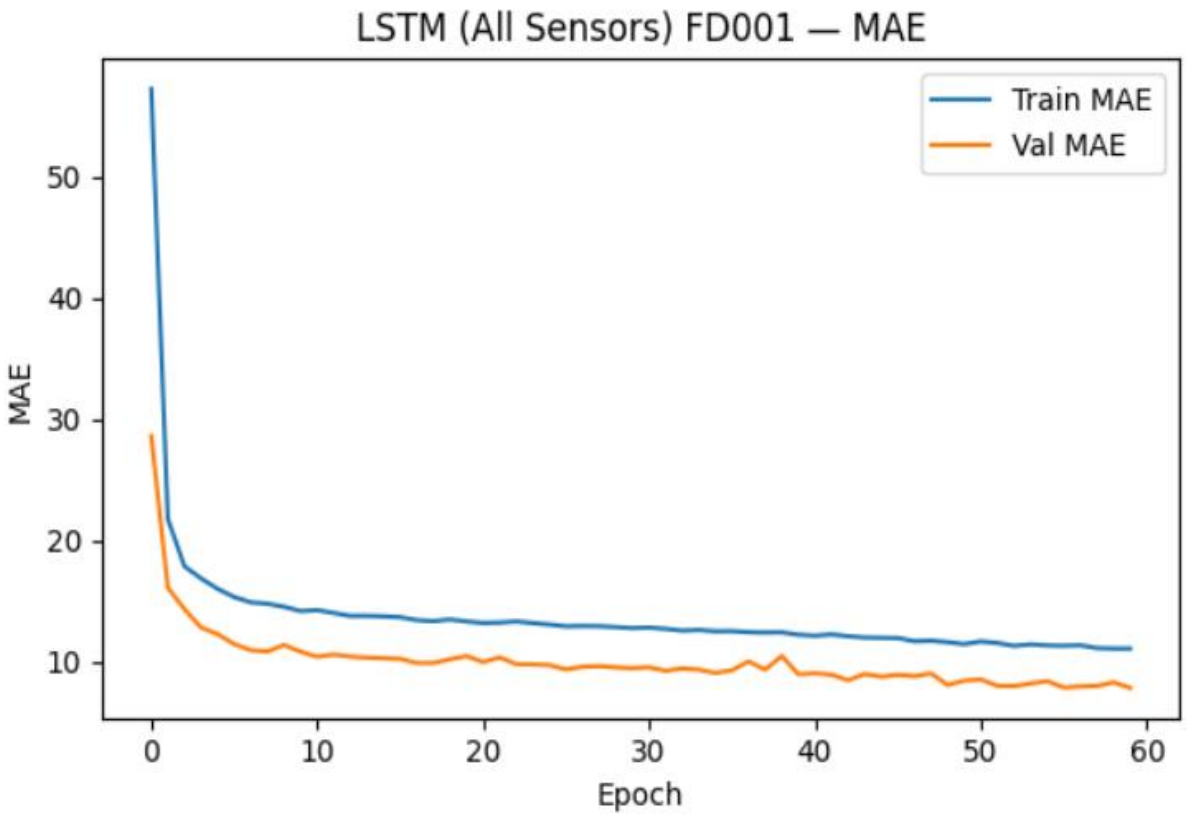
Model	Sensor Config	RMSE	PHM Score
CNN	All Sensors	19.671	521.88
CNN	Filtered	22.148	788.71
LSTM	All Sensors	16.125	516.24
LSTM	Filtered	16.134	558.31
CNN+BiLSTM	All Sensors	19.667	1389.12
CNN+BiLSTM	Filtered	19.205	1166.86

Ablation Study results(Filtered vs non filtered)-FD002

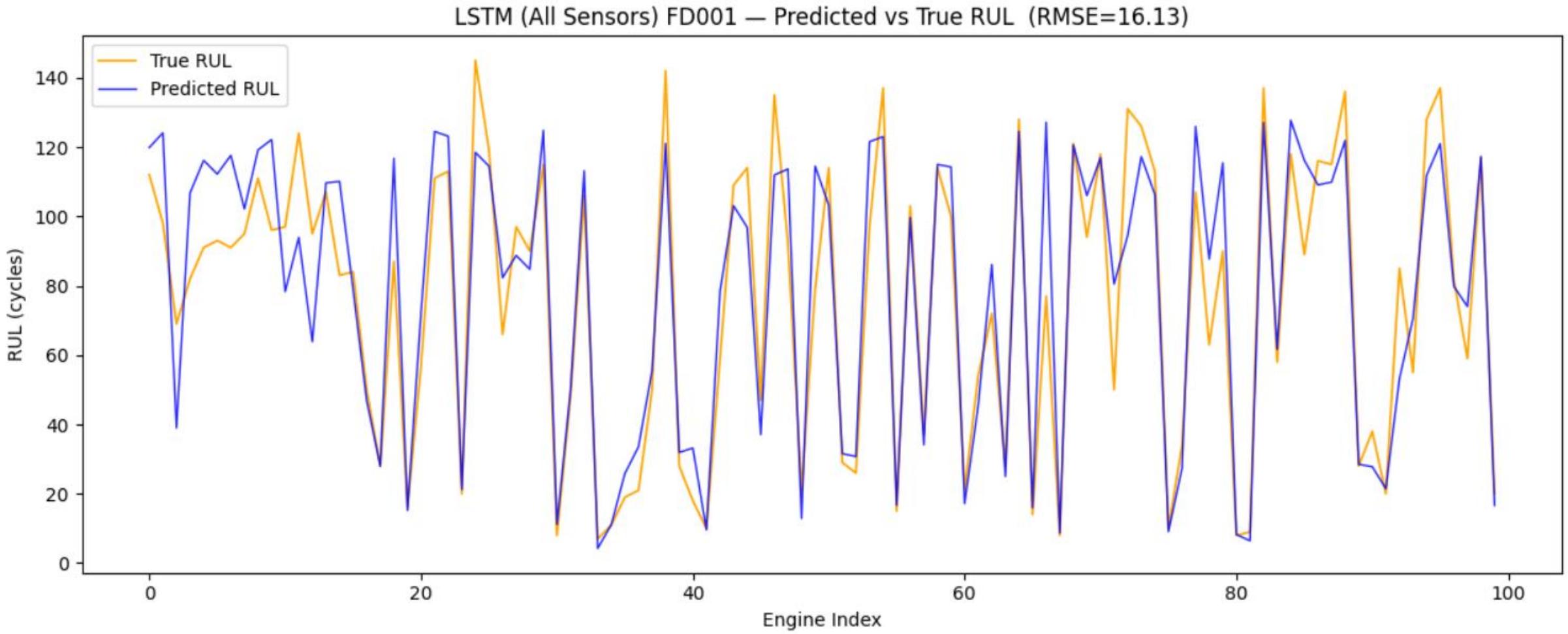
Model	Sensor Config	RMSE	PHM Score
CNN	All Sensors	30.498	17510.06
CNN	Filtered	32.585	28887.35
LSTM	All Sensors	28.924	21432.02
LSTM	Filtered	30.466	16901.39
CNN+BiLSTM	All Sensors	29.696	22501.43
CNN+BiLSTM	Filtered	31.803	30125.20

LSTM loss curves-FD001

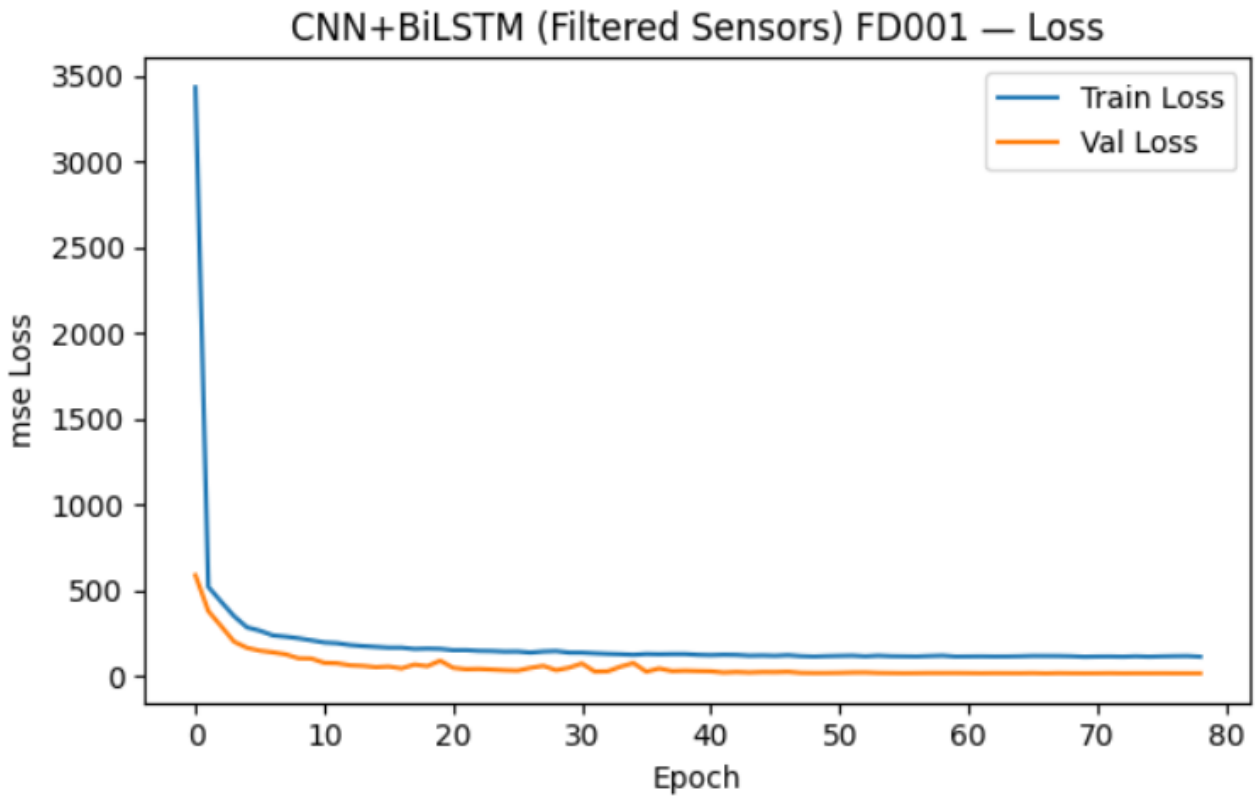


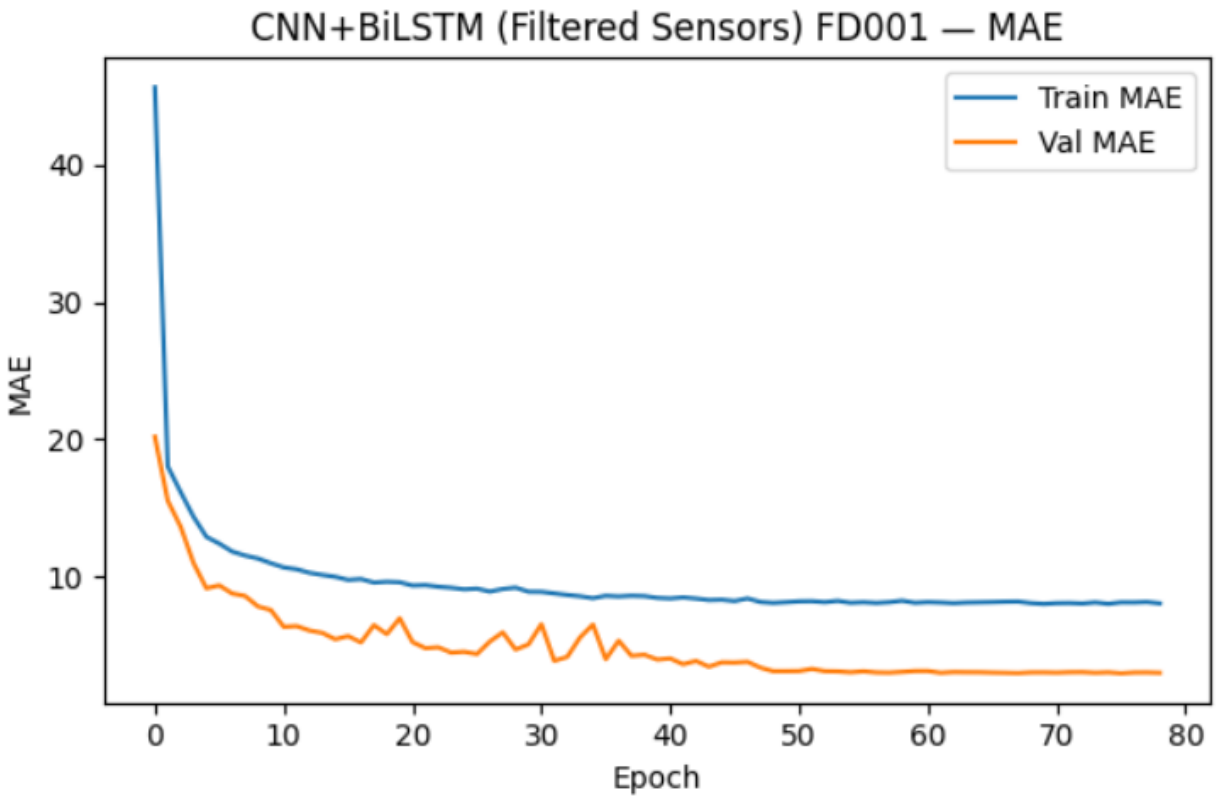


LSTM Predicted vs True RUL-FD001

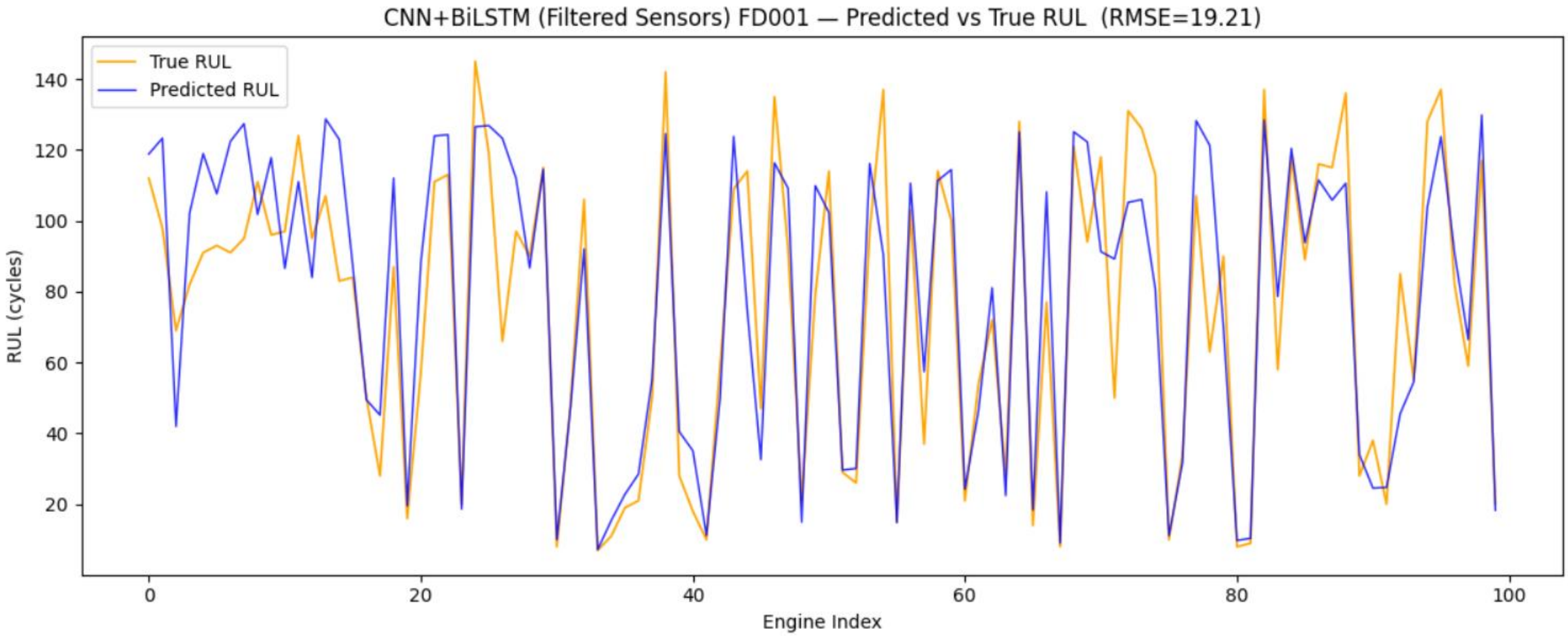


CNN+Bi-LSTM loss curves-FD001

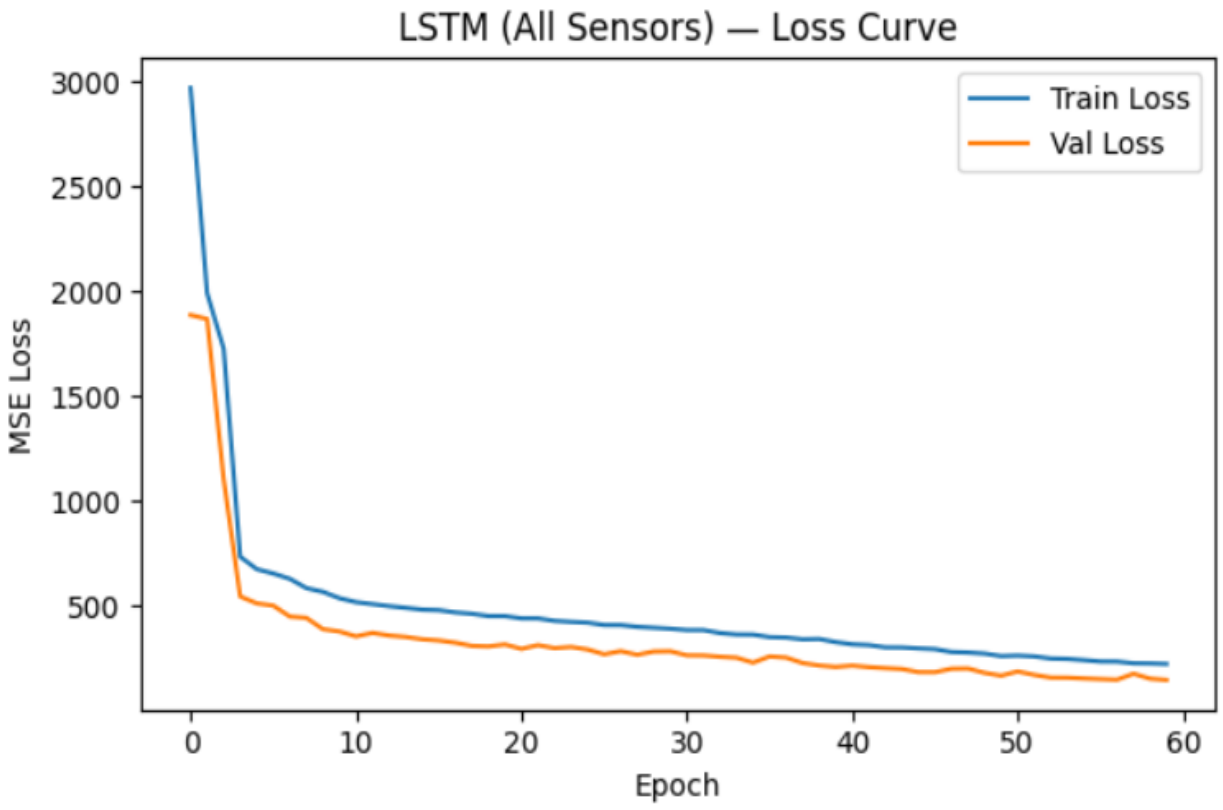


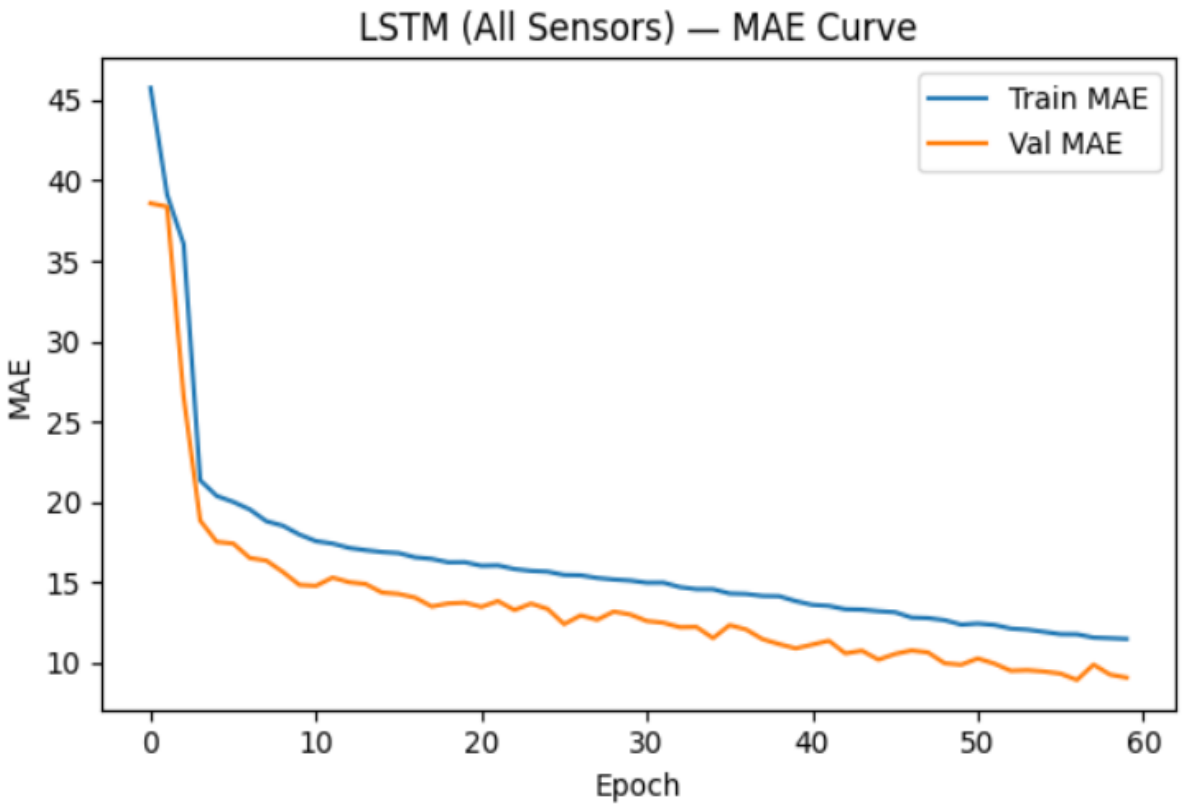


CNN+BiLSTM Predicted vs True RUL-FD001

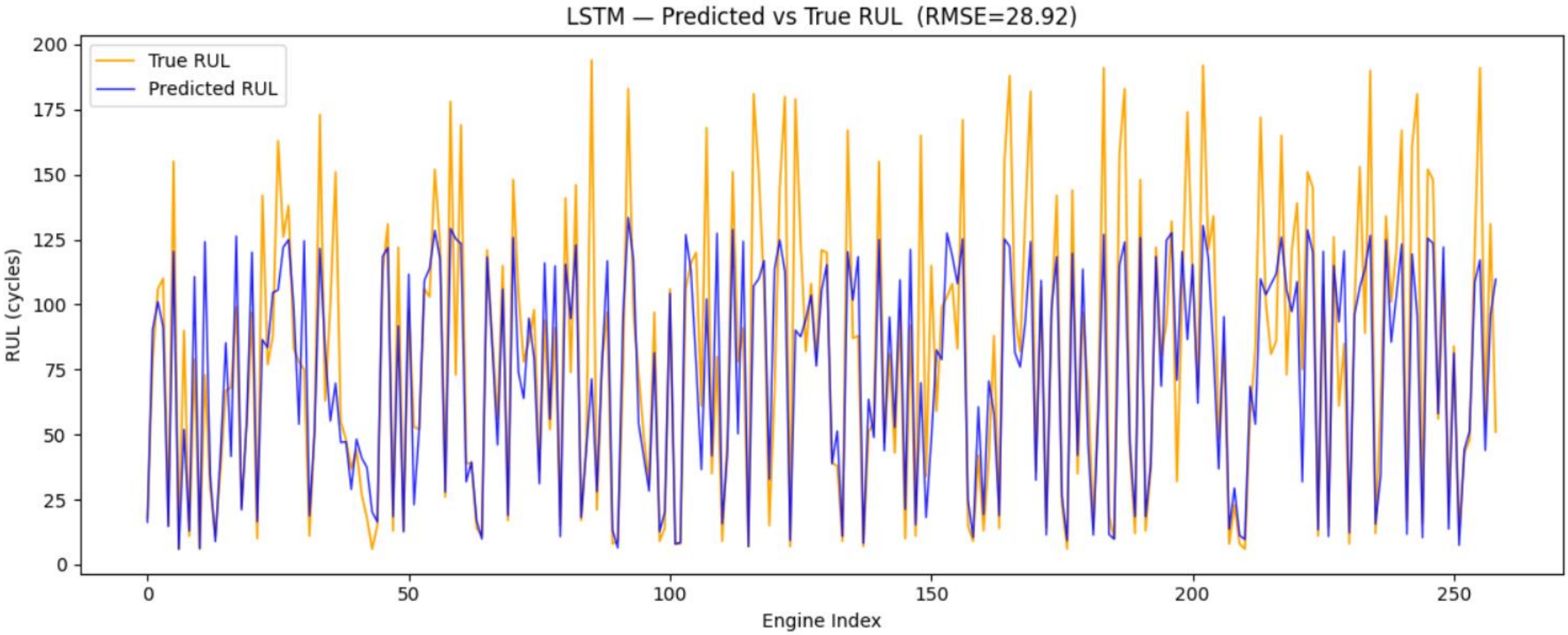


LSTM loss curves-FD002

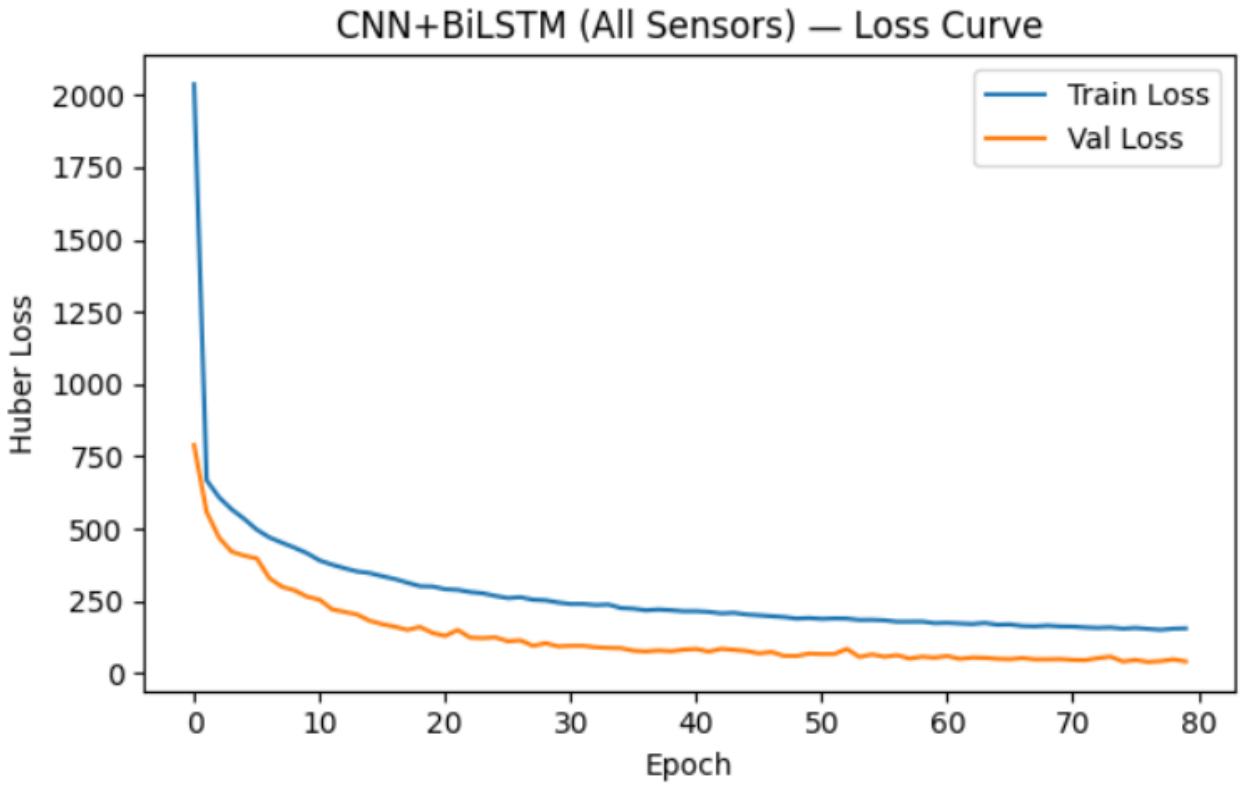


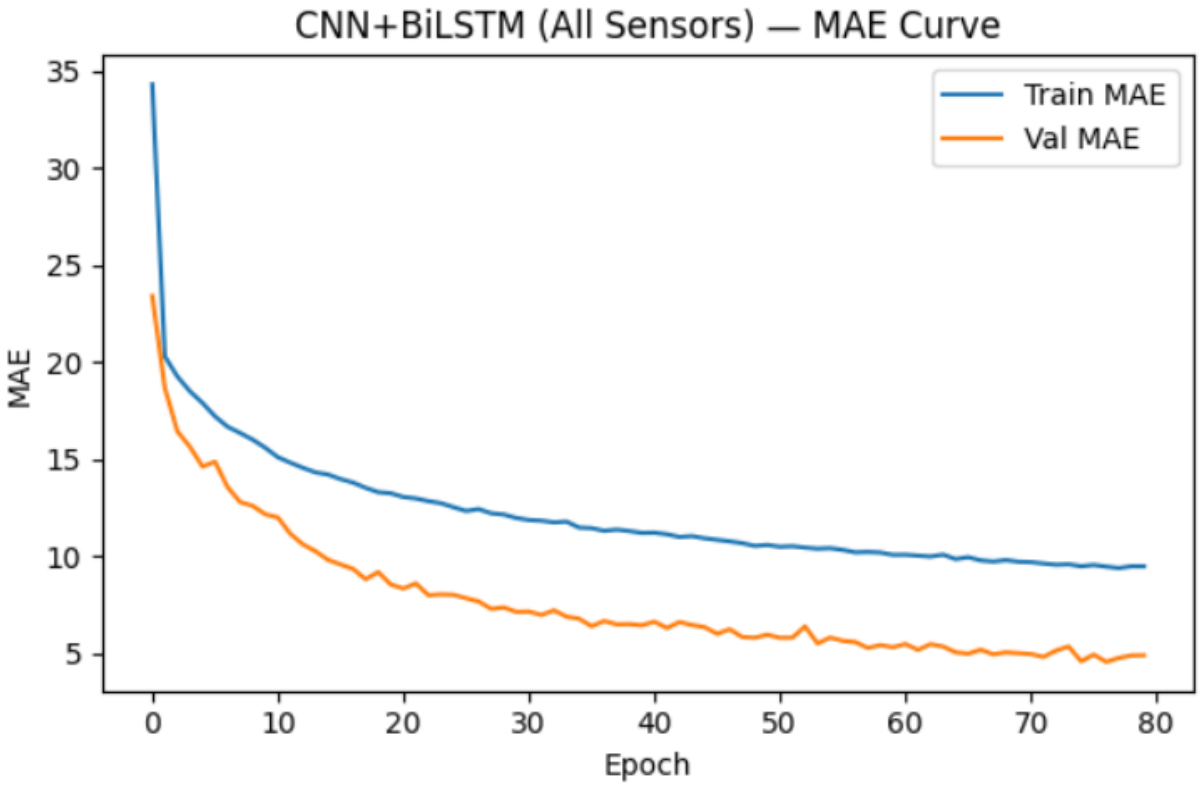


LSTM Predicted vs True RUL-FD002

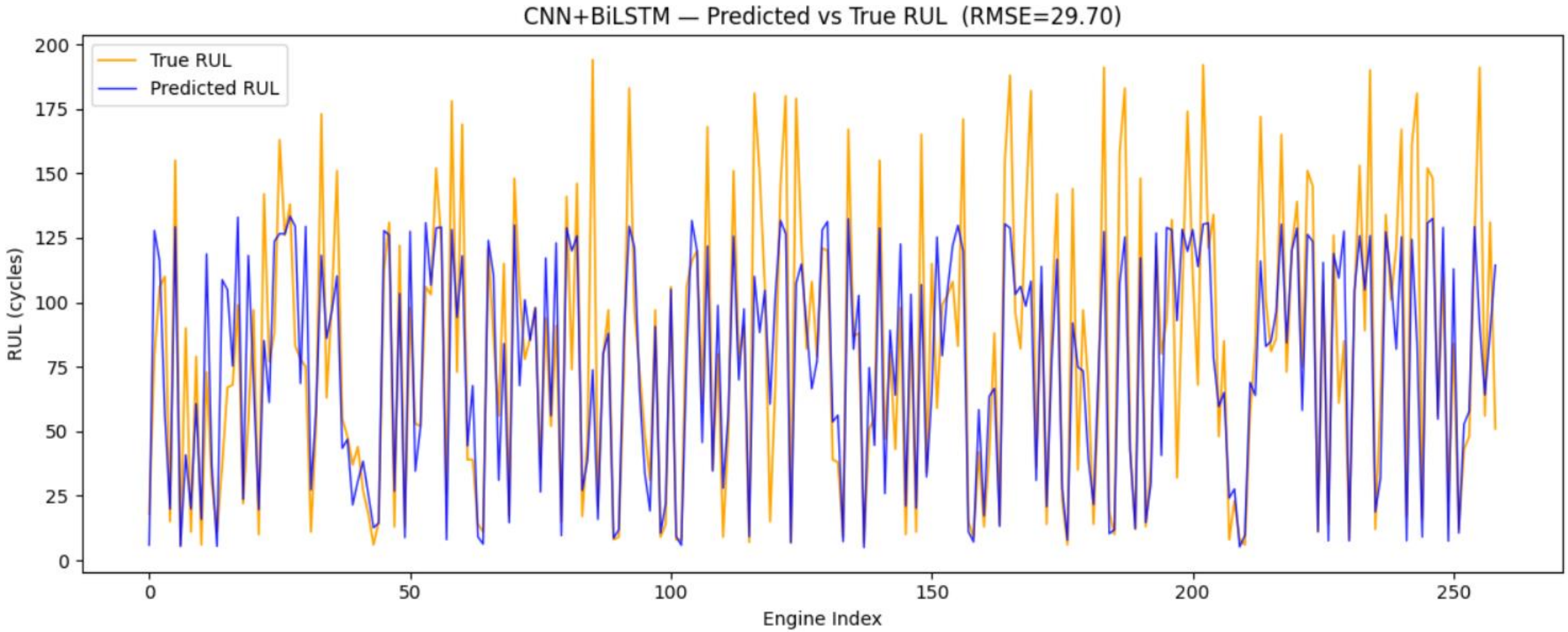


CNN+Bi-LSTM Loss curves-FD002

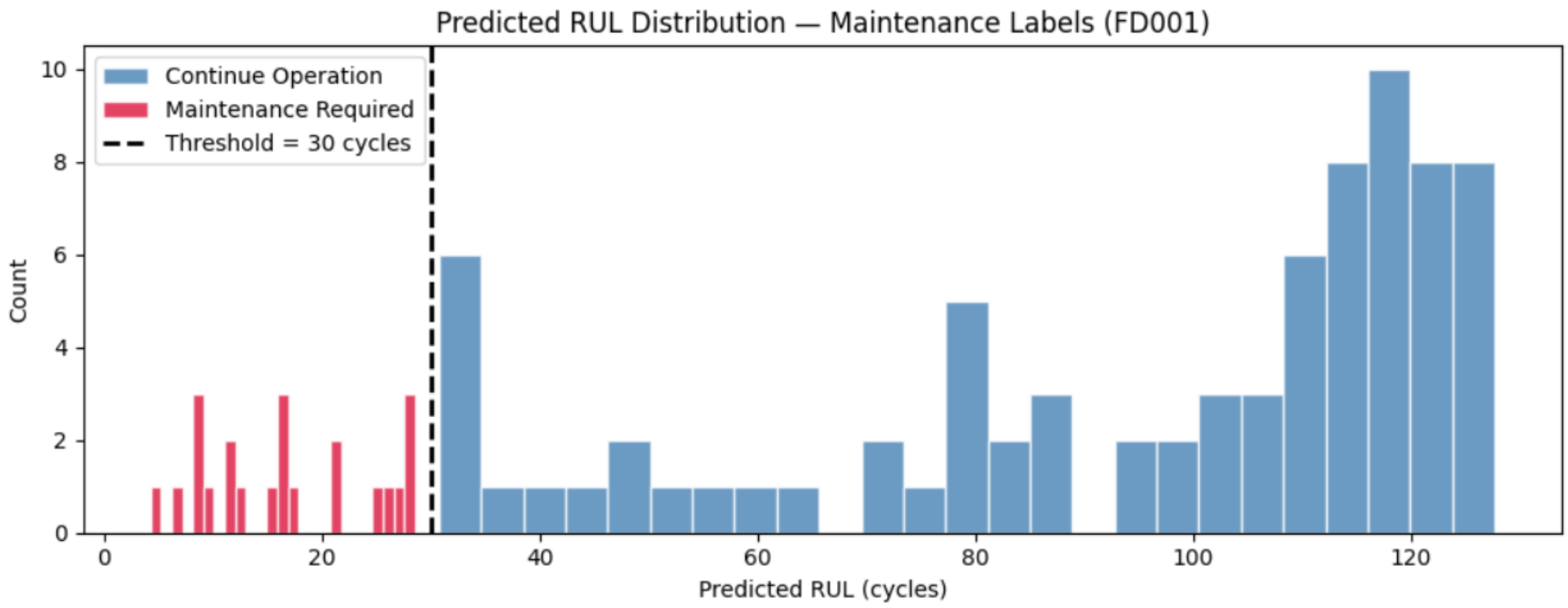




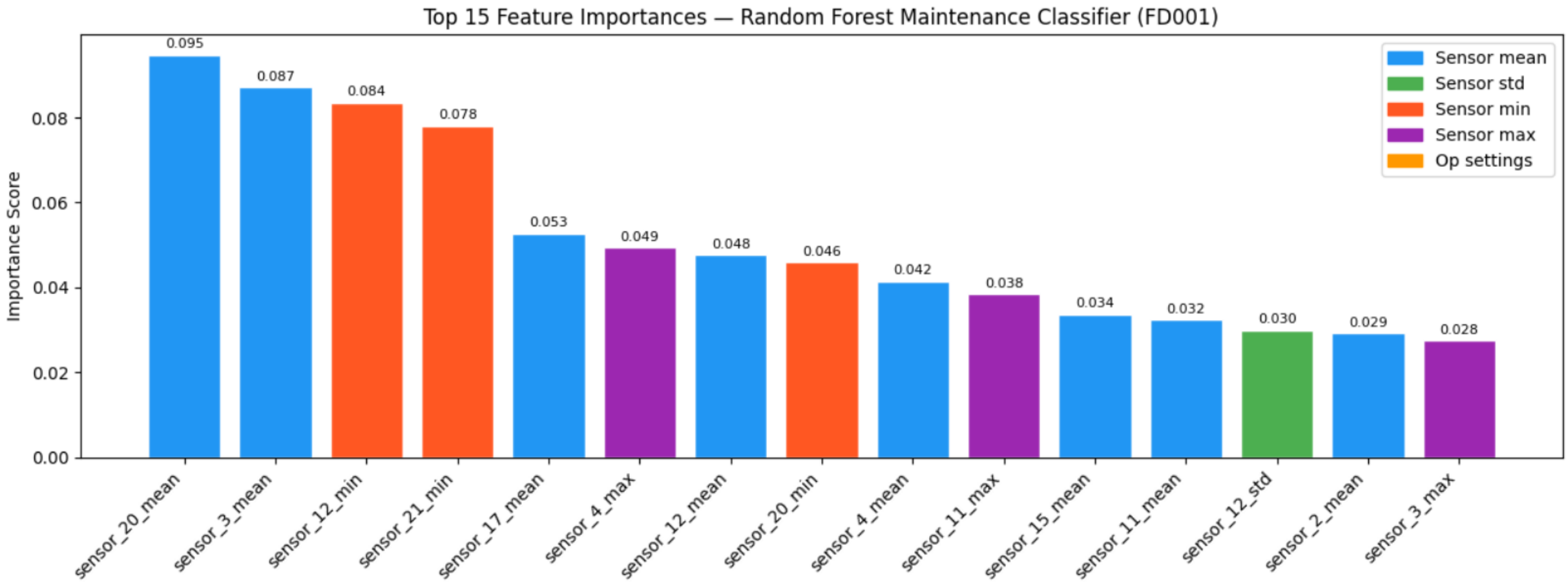
CNN+BiLSTM Predicted vs True RUL-FD002



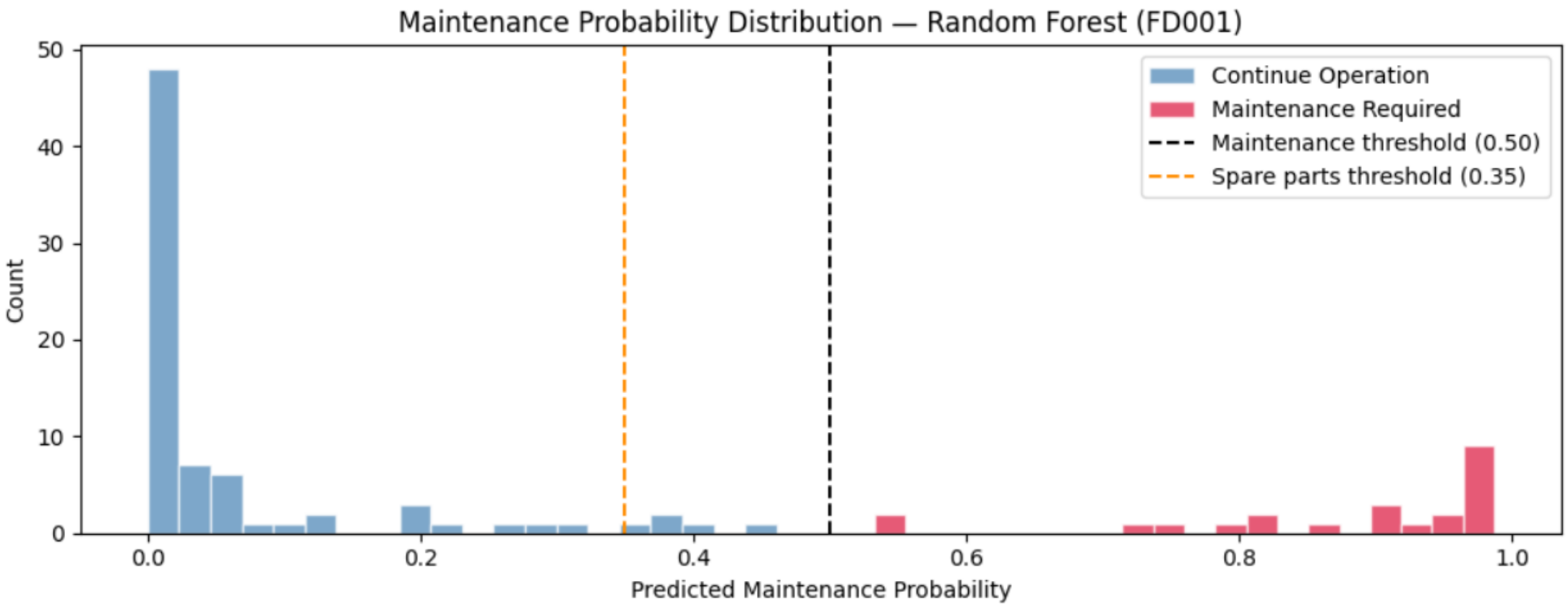
Maintenance Decision-Predicted RUL Distribution plot



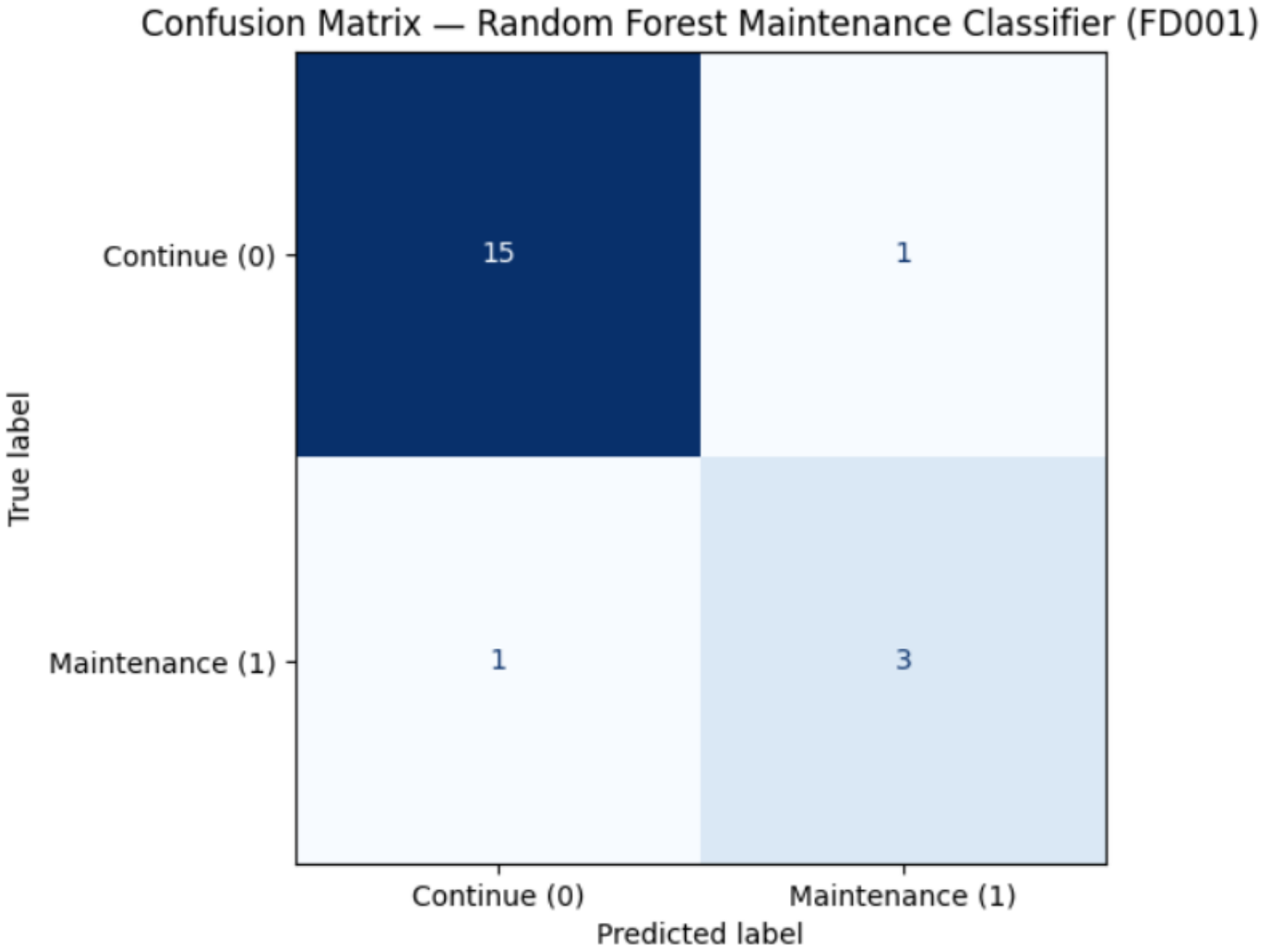
Feature Importance plot (Top 15)



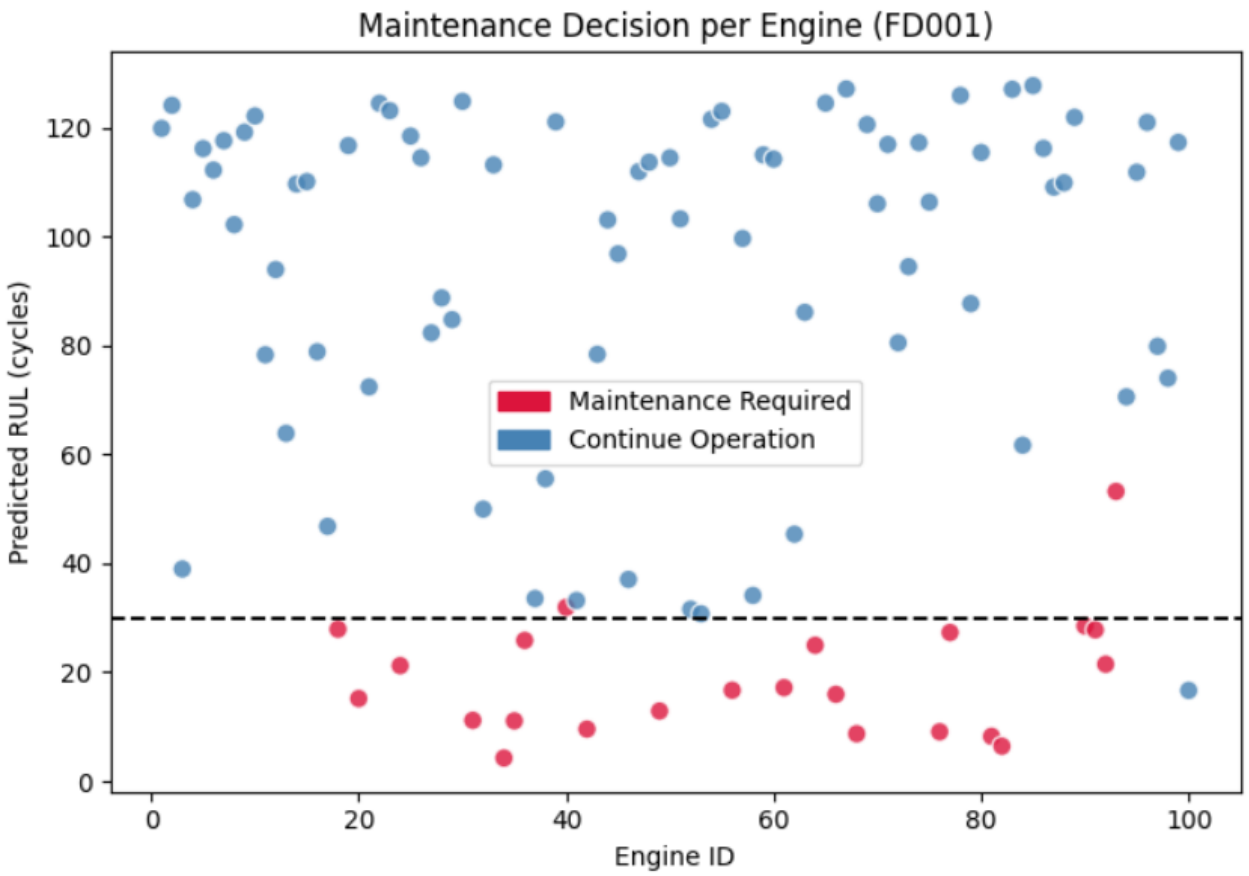
Data-Driven Maintenance Decision Framework — Label Generation and Classifier Confidence (FD001)

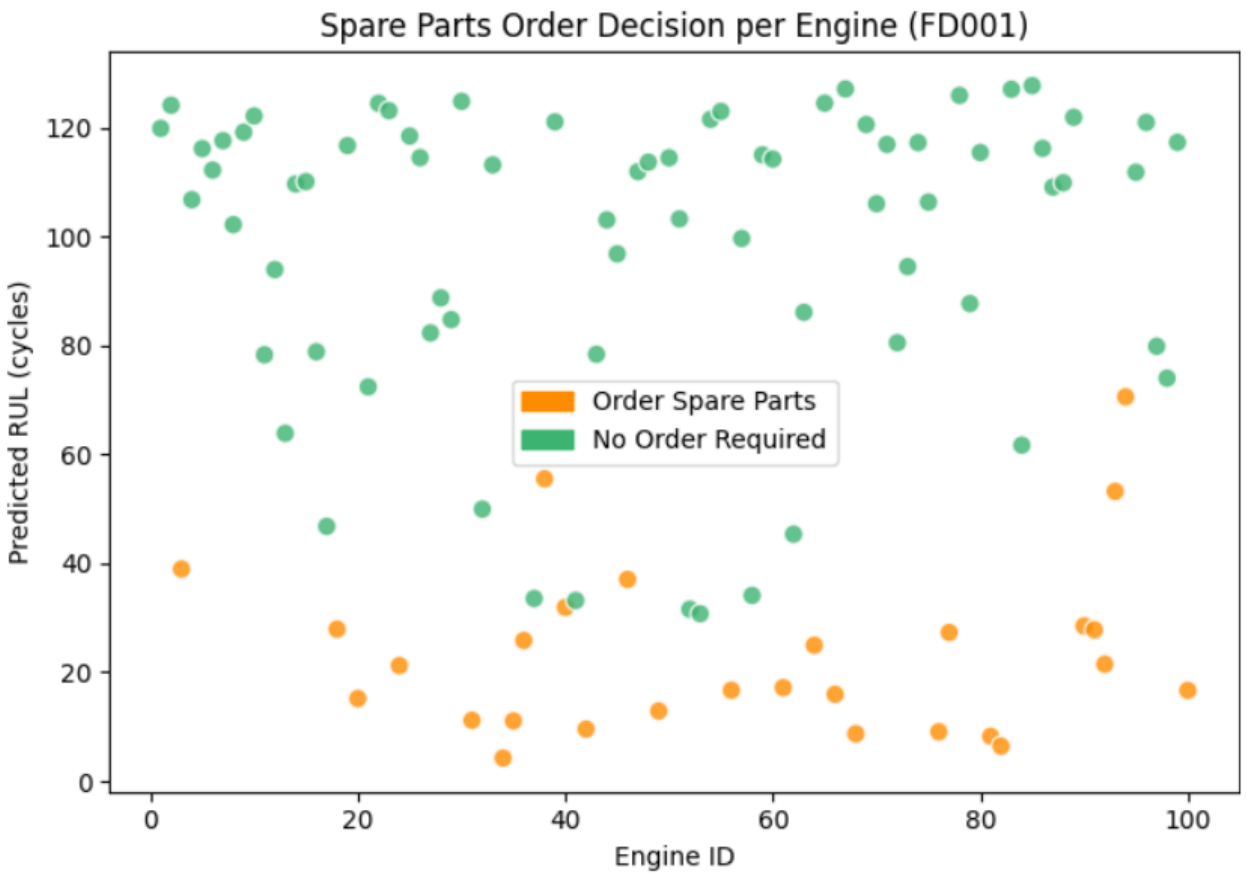


Confusion matrix



Maintenance Decision Scatter plot





Key Findings — RUL Prediction(FD001)

- LSTM with all 21 sensors achieved the best overall performance on FD001 with RMSE = 16.125 and MAE = 12.123, outperforming both CNN and CNN+BiLSTM across all sensor configurations.
- Sensor removal (filtered) did not improve performance on FD001 for any model — in fact CNN degraded significantly from RMSE 19.671 to 22.148 after filtering, indicating that the removed sensors still contained useful degradation information in a single-condition environment.
- LSTM showed remarkable stability across sensor configurations — RMSE changed by only 0.009 (16.125 vs 16.134) between all-sensor and filtered versions, confirming LSTM's robustness to irrelevant sensor noise on clean single-condition data
- On FD001, simpler sequential modelling (LSTM) outperforms hybrid architectures, suggesting that when the operating environment is stable, long-range temporal memory alone is sufficient for accurate RUL estimation

Key Findings — RUL Prediction(FD002)

- LSTM with all 21 sensors achieved the best RMSE (28.924) and MAE (19.960) on FD002, again outperforming all other model-sensor combinations.
- Unlike FD001, sensor removal consistently degraded performance on FD002 across all three models — CNN RMSE increased from 30.498 to 32.585, LSTM from 28.924 to 30.466, and CNN+BiLSTM from 29.696 to 31.803 after filtering.
- All models show significantly higher RMSE on FD002 (28–33) compared to FD001 (16–22), directly quantifying the added difficulty of predicting RUL across 6 operating conditions vs 1 — a 75–80% increase in error

Key Findings — Data-Driven Maintenance Decision Framework

- The Random Forest classifier successfully learned maintenance-critical patterns from temporal sensor statistics (mean, std, min, max over the 30-cycle window) with a accuracy of 90% and ROC-AUC score of 0.9688.
- Feature importance analysis revealed that specific degradation-sensitive sensors dominated the maintenance decision — the classifier autonomously identified which sensors are most predictive of imminent failure rather than relying on expert-defined rules
- The probability-based dual-threshold approach (maintenance at ≥ 0.50, spare parts at ≥ 0.35) produced differentiated decisions across the 100 test engines — engines with high degradation received early spare parts orders before crossing the critical maintenance threshold, demonstrating proactive scheduling behaviour.
- Replacing the static gamma distribution rule-based threshold with a data-driven Random Forest classifier eliminated the dependency on manually tuned parameters (λ, k, mission cycle Δt), making the maintenance schedule adaptive to actual observed degradation rather than assumed failure distributions.

Thank You